CARLYN LEE

626-419-6597 carlyn.lee@gmail.com

Accomplished Software Engineer specializing in scalable solutions, system reproducibility, and advanced visualization tools for high-performance computing. Experienced in interdisciplinary collaboration, complex algorithms, and system optimization with: *C/C++, Python, Java, MATLAB, Ansys STK*

EXPERIENCE

Oligo Space Jan 2025 - Present
Head of Software Engineering

- Architected Zenith, an end-to-end spacecraft design automation framework connecting payload requirements to complete satellite configurations.
- Developed automated pipelines for S-band, X-band, and UHF link budget analysis, regulatory filings (FCC Part 25 / ITU Appendix B & C), and integration with commercial ground-station networks.
- Integrated onboard networking components (SatCat5 FPGA switch, Arty A7-100T, Raspberry Pi-based data paths) for payload developer kits and flight-software interfaces.
- Designed internal tooling to support rapid spacecraft iteration, GMAT-based orbit analysis, and reinforcement-learning prototypes for structural mass-model optimization.

NASA Jet Propulsion Laboratory, California Institute of Technology Aug 2012 - Nov 2024
Applications Software Engineer *Pasadena, CA*

- **Tools Development:** Built deep space mission telecom simulation and modeling tools. Led and mentored remote interns on mission-critical systems. Automated operations enabling three Deep Space Network sites to manage the network during day shifts instead of 24/7.
- **Mission Operations:** Perseverance Rover telecom chair, responsible for rover communications health using data from Mars. Delivered tools reducing operation response-time from several hours to 20 minutes.
- **Supercomputing:** Developed Lunar terrain mask algorithms for Shaheen II supercomputer, enabling 10m-resolution link coverage maps. Parallelized Deep Space Optical Communications simulations, achieving bit error rate of 10e-8, a 1000x improvement.
- **Fieldwork:** Reliable multi-agent data transfer processes at transport layer. Field tested autonomous systems in maritime and subterranean environments, contributing to 1st place in DARPA's Urban Circuit Challenge.
- **Verification & Validation:** Developed automated tools and test procedures for avionics flight software, executing tests on flight hardware.

EXTRACURRICULAR, VOLUNTEER & PROFESSIONAL AFFILIATIONS

2019 - Present Global Network Advancement (GNA-G) Data Intensive Sciences Working Group, enabling high-throughput data transfers for Caltech, SCinet demonstration at Supercomputing 24-25.
2018 - Present SoCal Linux Expo Volunteer, deploying on-site network infrastructure and AV systems.
2014 - Present Interplanetary Small Satellite Conference committee member, managing budgets and logistics for 100+ industry leaders.

EDUCATION

California State University, Fullerton
M.S. Computer Science *August 2012*
B.S. Computer Science, Minor in Mathematics *July 2011*

PUBLICATIONS

K. Cheung; V. Vilnrotter; M. Sanchez-Net; C. Lee. High Data Rates from the Outer Solar System. IEEE Aerospace Conference. Big Sky, MT, USA. March 2025.

Vander Hook, J. V., Seto, W., Nguyen, V., Hasnain, Z., Lee, C.-A., Gallagher, L., Halpin-Chan, T., Varahamurthy, V., & Angulo, M. (2022). Swarms of Pirates: Red Team Exercises Using Autonomous High-Speed Maneuvering Surface Vessels. Field Robotics, 2, 872–909.

C. Lee, M. Shaikh, C.H. Lee and D. Michels. Lunar Terrain Coverage Analysis Data Delivery Workflow, AIAA 2021-4039. ASCEND 2021. November 2021.

A. Agha et al.. NeBula: Quest for Robotic Autonomy in Challenging Environments; TEAM CoSTAR at the DARPA Subterranean Challenge. Journal of Field Robotics, 2021.

C. Lee, H. Xie, C.H. Lee, D. Lyakhov, and D. Michels. In Silico Methods for Space System Analysis: Optical Link Coding Performance and Lunar Terrain Masks. In AIAA ASCEND, Las Vegas, NV, 16-18 Nov. 2020.

D. Abraham, B. MacNeal, D. Heckman, Y. Chen, J. Wu, K. Tran, A. Kwok and C. Lee. Recommendations Emerging from an Analysis of NASA's Deep Space Communications Capacity. In International Conference On Space Operations (SpaceOps 2018), Marseille, France, May 2018.

J. Lad, M. Johnston, D. Tran, D. Brown, K. Roffo, and C. Lee. Complexity-Based Link Assignment for NASA's Deep Space Network for Follow-the-Sun Operations. In International Conference On Space Operations (SpaceOps 2018), Marseille, France, May 2018.

K. Pinover, M. Johnston, C. Lee. Optimizing SmallSat Scheduling for NASA's Deep Space Network. In International Workshop on Planning and Scheduling for Space (IWPSS 2017). Pittsburgh, PA, June 2017.

D. Morabito, D. Kahan, K. Oudrhiri, and C. Lee. Cassini Downlink Ka-Band Carrier Signal Analysis. The Interplanetary Network Progress Report, Volume 42-208, February 15, 2017.

K. Cheung, D. Abraham, M. Sanchez-Net, K. Tran, C. Lee. Traffic modeling for Deep Space Network in the Human Exploration Era. In SpaceOps 2016 Conference, Daejeon, Korea, May 16-20, 2016.

M. Johnston, C. Lee, C. Lau, K. Cheung, M. Levesque, B. Carruth, A. Coffman, M. Wallace. Integrating space communication network capabilities via web portal technologies. In SpaceOps 2014 13th International Conference on Space Operations, Pasadena, California, May 5-9, 2014.

C. Lee, C.H. Lee. Cancer Screening Using Multi-modal Differential Principal Orthogonal Decomposition. In 2013 13th International Conference on Computational Science and Its Applications, Ho Chi Minh City, Vietnam, June 24-27 2013.

C. Lee, C.H. Lee. Extended Principal Orthogonal Decomposition Method for Cancer Screening. International Journal of Bioscience, Biochemistry and Bioinformatics vol. 2, no. 2, pp. 136-141, 2012.

C. Lee. Rest architecture for link analysis tools portal. NASA Undergraduate Student Research Program (USRP), Pasadena, California, August 2011.